

agentic Systems

✱ The primary goal of agentic systems *create controlled tool*

Using LLM systems

→ make LLM choose action instead of returning text:

call tools, query Database, Browse docs, write Files, send email -- etc

More autonomy means more Failure surface.

WorkFlows vs Agents

work Flow
First.

agent only
where decision
Flexibility is
needed

— use Deterministic over Flow when steps are known

— use agents when Paths must be chosen dynamically

architecture
Patterns

Path or model to use
Router, tool caller, Planner, Executor, Verifier
action
human approval gate
stop risky action before execution
checks outputs

State,
memory,
Failure

Agents require state to function

→ state may include the user goals, Plan, steps error, budget... etc

~~✗~~ Critically, memory must be controlled: unbounded memory

create confusion and security risks. Like:

tool misuse, infinite loop, state memory, prompt injection
hidden cost explosion and invalid answer